

SISSETON, SD, USA

WMO: 726519

Lat: 45.669N

Lon: 96.991W

Elev: 1161

StdP: 14.09

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 94993

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -13.7	(c) -9.2	(d) -20.4	(e) 1.9	(f) -12.4	(g) -16.4	(h) 2.3	(i) -9.0	(j) 35.0	(k) 18.9	(l) 32.5	(m) 20.0	(n) 9.1	(o) 320	(p) 0.654

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.4	(c) 90.2	(d) 72.6	(e) 87.3	(f) 71.4	(g) 84.3	(h) 69.6	(i) 76.5	(j) 86.1	(k) 74.4	(l) 83.3	(m) 72.5	(n) 81.1	(o) 14.2	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
73.3	129.4	81.9	71.6	122.0	80.6	69.6	113.8	78.4	40.8	86.4	38.8	83.2	36.9	80.8	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 29.7	(b) 26.5	(c) 24.1	(e) -22.0	(f) 95.8	(g) 5.3	(h) 3.3	(i) -25.8	(j) 98.1	(k) -28.9	(l) 100.0	(m) -31.8	(n) 101.9	(o) -35.7	(p) 104.2		
(4) 29.7	(5) 26.5	(6) 24.1	(7) -22.1	(8) 79.9	(9) 4.9	(10) 1.9	(11) -25.6	(12) 81.3	(13) -28.4	(14) 82.4	(15) -31.2	(16) 83.5	(17) -34.7	(18) 84.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	44.3	14.3	15.4	30.0	44.1	56.8	67.3	71.7	68.4	61.4	47.0	33.1	20.2
	DBStd	22.85	14.38	13.01	13.31	10.58	9.26	6.15	5.65	5.81	8.55	9.43	11.31	13.42
	HDD50	4592	1106	970	629	228	39	0	0	0	9	172	512	927
	HDD65	8124	1571	1390	1086	627	285	45	11	30	168	565	956	1390
	CDD50	2511	0	0	9	51	251	519	673	570	350	79	7	2
	CDD65	568	0	0	0	1	32	114	219	136	60	6	0	0
	CDH74	5855	0	0	4	45	442	1130	2193	1297	661	83	0	0
	CDH80	1881	0	0	0	10	149	341	766	384	208	23	0	0
Wind	WSAvg	10.8	12.0	11.8	11.9	12.4	12.0	9.9	8.2	7.8	9.5	11.0	11.5	11.8
Precipitation	PrecAvg	25.6	0.7	0.8	1.3	2.4	3.0	4.4	3.5	2.6	2.8	2.4	1.0	0.7
	PrecMax	30.2	2.1	1.7	3.9	5.2	5.5	8.2	7.3	4.5	7.0	7.1	3.9	2.0
	PrecMin	19.8	0.0	0.1	0.0	0.2	1.2	1.7	1.0	0.6	0.1	0.3	0.0	0.2
	PrecStd	3.5	0.6	0.5	0.8	1.2	1.1	1.8	1.6	1.2	1.6	1.9	1.0	0.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.0	52.7	70.4	79.3	90.3	91.8	95.3	91.5	91.1	83.2	66.4	60.6
		MCWB	38.1	46.7	55.7	58.3	67.0	71.7	76.5	75.2	71.7	63.2	53.1	40.4
	2%	DB	41.7	44.0	62.0	72.6	84.2	87.3	90.2	87.9	85.8	74.1	60.1	48.0
		MCWB	35.4	38.1	51.5	54.6	63.9	70.0	75.1	72.9	69.7	58.3	49.0	38.5
	5%	DB	38.2	38.0	54.6	67.2	78.9	83.8	87.1	84.0	81.4	68.0	54.3	42.2
		MCWB	33.9	34.4	46.7	52.9	61.2	67.5	73.1	71.1	66.5	55.5	45.3	35.6
	10%	DB	34.5	34.3	48.0	62.4	73.4	80.8	84.2	81.2	77.3	62.9	48.8	37.2
		MCWB	31.8	31.6	42.6	49.9	58.8	67.0	71.3	69.2	64.8	52.6	41.5	32.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.4	45.9	59.1	60.8	70.2	76.8	80.1	77.3	74.7	64.6	55.1	43.1
		MCDB	46.1	53.6	65.0	74.4	84.8	85.6	88.3	87.4	85.1	77.4	64.5	52.5
	2%	WB	36.0	38.3	52.4	56.9	66.5	73.8	77.4	75.2	71.6	60.4	50.1	38.5
		MCDB	40.4	43.2	60.4	69.5	78.3	82.8	87.3	84.4	82.1	71.5	58.5	47.4
	5%	WB	34.1	34.7	47.1	53.6	64.0	70.9	75.3	73.1	69.0	56.8	45.8	35.6
		MCDB	37.8	38.0	54.4	63.9	74.3	80.2	84.6	81.3	78.3	66.6	53.6	41.4
	10%	WB	32.0	31.7	42.3	50.9	60.9	68.5	73.3	71.0	66.3	53.3	42.0	33.1
		MCDB	35.0	34.3	48.0	60.5	70.9	77.4	81.5	78.9	74.9	61.3	47.8	37.3
Mean Daily Temperature Range	5% DB	MDBR	16.8	17.1	18.0	21.4	22.1	20.4	21.4	21.8	23.0	19.8	17.8	15.3
		MCDBR	20.0	19.8	28.2	31.7	30.1	25.6	24.1	24.6	27.5	30.0	25.7	20.7
		MCWBR	15.8	15.4	18.2	17.3	14.6	12.5	12.3	12.4	13.3	16.5	16.1	14.4
	5% WB	MCDBR	18.6	19.5	27.0	27.6	25.5	21.5	21.7	22.3	24.1	26.2	25.3	19.9
		MCWBR	15.0	15.4	18.0	16.3	13.8	12.6	12.3	12.5	13.5	15.6	16.5	14.6
Clear-Sky Solar Irradiance	taub	0.277	0.288	0.310	0.356	0.377	0.388	0.388	0.395	0.359	0.322	0.287	0.266	
	taud	2.373	2.371	2.425	2.333	2.354	2.369	2.366	2.332	2.415	2.490	2.493	2.426	
	Ebn at Noon	265	285	293	286	282	279	277	271	272	269	261	258	
	Edh at Noon	23	28	31	37	38	38	37	37	31	25	21	20	
All-Sky Solar Radiation	RadAvg	551	899	1211	1541	1753	1970	2075	1739	1334	845	547	402	
	RadStd	59	96	109	115	121	115	82	149	89	112	59	41	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page